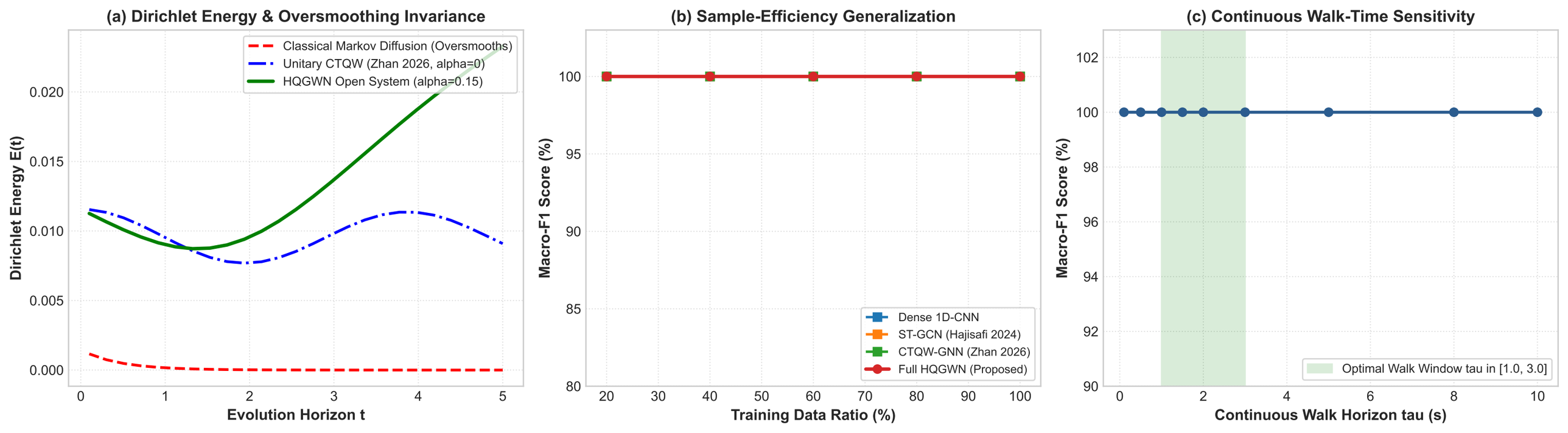
(a) Dirichlet Energy & Oversmoothing Invariance
Classical Markov Diffusion (Oversmooths)
Unitary CTQW (Zhan 2026, alpha=0)
HQGWN Open System (alpha=0.15)
Dirichlet Energy E(t)
0.020
0.015
0.010
0.005
0.000
0
1
2
3
4
5
Evolution Horizon t
(b) Sample-Efficiency Generalization
Macro-F1 Score (%)
100
95
90
85
80
20
30
40
50
60
70
80
90
100
Training Data Ratio (%)
Dense 1D-CNN
ST-GCN (Hajisafi 2024)
CTQW-GNN (Zhan 2026)
Full HQGWN (Proposed)
(c) Continuous Walk-Time Sensitivity
Macro-F1 Score (%)
102
100
98
96
94
92
90
0
2
4
6
8
10
Continuous Walk Horizon tau (s)
Optimal Walk Window tau in [1.0, 3.0]